East West University
Department of Computer Science and Engineering
Course Outline of CSE110
Summer 2025

Course Information

Course: CSE110 Object Oriented Programming (Section 3)

Credit and Teaching Scheme:

	Theory	Laboratory	Total
Credits	3.0	1.5	4.5
Contact Hours	3 Hours/Week for 13 Weeks	3 Hours/Week for 13 Weeks	6 Hours/Week for 13 Weeks

Prerequisite: CSE106 Discrete Mathematics

Instructor Information

Instructor: Dr. Mohammad Salah Uddin
Associate Professor, Department of Computer Science and Engineering
Office: Room # 648
Tel. No.: N/A
E-mail: uddin@ewubd.edu

Course Repository: www.banglarbot.com/oop
GTA: Golam Kibria

Class and Office Hour:

Day	08:30-10:00	10:10-11:40	11:50-01:20	01:30-03:00	03:10-04:40	04:50-06:20	06:20-07:50
Sunday				Office Hour	CSE106(4) Room#432	CSE227(1) LAB Room#637	CSE227(1) LAB Room#637
Monday	CSE106(5) Room#433	Office Hour					
Tuesday	Office Hour	CSE227(1) Room#FUB-104	Office Hour	CSE110(3) Room#220			
Wednesday	CSE106(5) Room#433	CSE110(3) LAB Room#637	CSE110(3) LAB Room#637				
Thursday	Office Hour	CSE227(1) Room#FUB-104	Office Hour	CSE110(3) Room#220	CSE106(4) Room#432		

Course Objective

This course presents a conceptual and practical introduction to object-oriented programming (OOP). The course will cover general principles of programming in object-oriented frameworks to enhance transferable skills, such as programming, designing, and problem-solving skills. This course introduces object-oriented concepts and develops OOP programs which provides solutions to real-world object-oriented problems. Java is primarily chosen as the programming language in this course. Knowledge of this course will be needed as prerequisite knowledge for CSE207 Data Structures.

Course Outcomes (COs)

After completion of this course, students will be able to:

CO1	**Understand** and **apply** the fundamental principles of object-oriented programming in the target language to solve problems using an object-oriented approach.
CO2	**Use** the principles of inheritance, polymorphism, abstraction, and interfaces to develop reusable and modular code.
CO3	**Apply** exceptions, generics, inner classes, and **perform** file I/O operations to develop robust and data-driven applications.
CO4	**Choose** appropriate tools, **perform** and **demonstrate** skills, and write a report to design, build, and test interactive applications that incorporate a graphical user interface (GUI) with the option of using multithreading and database connectivity in the target language to solve real-life complex problems.

Course Topics, Teaching-Learning Methods, and Assessment Scheme

Course Topic	**Teaching-Learning Method**	**CO**	**Mark of Cognitive Learning Levels**		**Mark of COs**	**Exam (Mark)**
			C2	C3		
Principles of Object-Oriented Programming and Basics of Elementary Programming in Java (conditional branching, looping, methods, and arrays)	Lecture, Class Discussion, Discussion Outside Class with Instructor/ Teaching Assistant	**CO1**	2.5	2.5	5	**Mid Term Exam (25)**

Introduction to Classes and Objects (Classes, Objects, Instance variables and instance methods, Constructors) **[Class Test 01 will be conducted on topics under CO1]**	Do		5	5	10	
Inheritance, Polymorphism in OOP (super class, sub class, multiple-level inheritance, late binding)	Do	**CO2**		10	10	
Abstract Class and Interfaces **[Class Test 02 will be conducted on topics under CO2]**	Do	**CO2**		10	10	**Final Exam (30)**
Exception Handling in OOP	Do	**CO3**		5	5	
File handling using Text and Binary I/O	Do			5	5	
Implementation of Collections, Generics, and Inner Classes **[Class Test 01 will be conducted on topics under CO3]**	Do			10	10	

Lab Exercises

Experiment	**Teaching-Learning Method**	**CO**	**Marks of Cognitive Level**	**Mark of Psychomotor Level**		**Mark of Affective Level**	**Mark of COs**
			C3	P2	P3	A2	
Java Basics of Elementary Programming, Conditional Statements	Lab Experiment and Result Analysis and Discussion with Instructor, Post-Lab Report	CO4					
Looping, Nested Looping, Arrays Java Methods and library functions	Do	CO4					
Designing and Implementing simple Classes and	Do	CO4					

Objects, Arrays of Objects etc.							
Implementing associations of Classes, and Inner class	Do	CO4					
Designing and Implementing Inheritance and Polymorphism	Do	CO4					
Designing and Implementing Abstract Class and Interfaces	Do	CO4					
Understanding and Implementing Exceptions and File management, Collections, Generics	Do	CO4					
GUI using JavaFX	Do	CO4					
Lab Experiments		CO4	3	2.5	2.5	2	10
Lab Final (Exam)	Individual Exam	CO4	3	2.5	2.5	2	10
Total			**6**	**5**	**5**	**4**	**20**

Mini Project

Mini Project	Teaching-Learning Method	CO	Mark of Cognitive Levels		Mark of Psychomotor Levels		Mark of Affective Levels	Mark of COs
			C3	C4	P2	P3	A2	
Mini Project including Report and Presentation	Moderately complex Project with report writing, and oral/poster presentation	CO4	**2**	**2**	**2**	**2**	**2**	**10**

Overall Assessment Scheme

	COs				Assessment Area Mark
Assessment Area	**CO1**	**CO2**	**CO3**	**CO4**	
Class Participation/Test/Quizzes	5	5	5		15
Mid Semester Assessment	15	10			25

Final Exam		10	20		30
Lab Performance				10	10
Lab Final				10	10
Mini Project Demonstration and Viva				10	10
Total Mark	**20**	**25**	**25**	**30**	**100**

Teaching Materials/Equipment

Text Book:

- **Introduction to Java Programming by Daniel Liang**
- **Herbert Schildt, *Java: The Complete Reference*, 11th edition, McGraw-Hill Education (2023)**

Reference Book:

- Paul Deitel, Harvey Deitel, Java™ How to Program Early Objects, 11th edition
- Walter Savitch, *Absolute Java*, Pearson (5th edition)
- Bert Bates and Kathy Sierra, *Head First Java*, O'Reilly Media (2nd edition)

Software/Tools:

- Java Development Kit (JDK 1.8) https://www.oracle.com/technetwork/java/javase/downloads/jdk8-downloads-2133151.html
- Any Integrated Development Environment (IDE) supporting Java preferably NetBeans or Eclipse https://netbeans.apache.org/download/index.html, https://www.eclipse.org/downloads/
- Android Studio

Grading System

Marks (%)	Letter Grade	Grade Point	Marks (%)	Letter Grade	Grade Point
80-100	A+	4.00	55-59	B-	2.75
75-79	A	3.75	50-54	C+	2.5
70-74	A-	3.5	45-49	C	2.25
65-69	B+	3.25	40-44	C-	2
60-64	B	3.00	Below 40	F	0.00

Exam Dates

As per the schedule provided by the department and the university.

Section	Class Slot	Mid Semester	Final
4	SR	31 July 2025	11 September 2025
5	MW	30 July 2025	10 September 2025

Academic Code of Conduct

Academic Integrity:
Any form of cheating, plagiarism, personification, falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offence under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

For details on Student Ethics and Academic Discipline, please visit https://www.ewubd.edu/student-rules-regulation

Special Instructions:

- Students are expected to attend all classes and examinations. A student MUST have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter into the classroom after 10 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss any exam, the student MUST get approval of makeup exam by written application to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason of missing the exam have to be presented with the application.
- For **final exam**, there will be NO makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the final exam, the student MUST get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor**.
- All mobile phones MUST be turned to silent mode during class and exam period.
- There is **zero tolerance for cheating** in exam. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university**.